

## QUINCY, IL, USA

WMO: 724430

Lat: 39.937N

Lon: 91.192W

Elev: 234

StdP: 98.54

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 93989

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -18.0      | -15.0 | -22.5                         | 0.5 | -17.5 | -19.9 | 0.7 | -14.2 | 12.6                  | -3.2 | 11.6 | -3.8 | 4.2                   | 310  | 0.526 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.4                   | 33.9            | 24.7 | 32.5 | 24.1 | 31.1 | 23.4 | 26.1                | 31.7 | 25.3 | 30.7 | 24.4 | 29.4 | 4.5                  | 220  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.5                            | 20.0 | 29.6 | 23.6 | 19.0 | 28.7 | 22.8 | 18.0 | 27.6 | 82.3          | 32.0 | 78.4 | 30.6 | 74.8 | 29.4 |                   |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 10.9              | 9.3  | 8.4 | -21.8                      | 36.4 | 3.1                | 1.9 | -24.0                                              | 37.8 | -25.9      | 38.8 | -27.6      | 39.9 | -29.9      | 41.2 |    |    |
|                   |      |     | -22.1                      | 27.4 | 3.0                | 1.0 | -24.3                                              | 28.1 | -26.1      | 28.7 | -27.8      | 29.3 | -30.0      | 30.0 |    |    |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 11.8   | -3.0  | -0.5  | 5.6   | 11.9  | 17.6  | 22.7  | 24.6  | 23.8  | 19.7  | 13.0  | 6.0   | -0.2 |
|                                                                               | DBStd       | 10.96  | 6.86  | 6.78  | 6.35  | 5.35  | 4.66  | 3.50  | 3.17  | 2.98  | 4.38  | 5.33  | 5.73  | 6.17 |
|                                                                               | HDD10.0     | 1406   | 404   | 298   | 167   | 39    | 3     | 0     | 0     | 0     | 1     | 29    | 146   | 319  |
|                                                                               | HDD18.3     | 3027   | 662   | 527   | 397   | 201   | 72    | 7     | 1     | 1     | 36    | 179   | 371   | 574  |
|                                                                               | CDD10.0     | 2074   | 1     | 5     | 30    | 97    | 238   | 380   | 453   | 427   | 291   | 123   | 26    | 3    |
|                                                                               | CDD18.3     | 652    | 0     | 0     | 1     | 9     | 49    | 137   | 196   | 169   | 76    | 14    | 1     | 0    |
|                                                                               | CDH23.3     | 5913   | 0     | 0     | 8     | 83    | 385   | 1209  | 1947  | 1487  | 677   | 114   | 2     | 0    |
|                                                                               | CDH26.7     | 2152   | 0     | 0     | 0     | 12    | 93    | 433   | 802   | 554   | 234   | 24    | 0     | 0    |
| Wind                                                                          | WSAvg       | 4.1    | 4.7   | 4.7   | 4.9   | 4.9   | 4.1   | 3.6   | 3.1   | 3.0   | 3.3   | 4.0   | 4.5   | 4.4  |
| Precipitation                                                                 | PrecAvg     | 993    | 37    | 42    | 75    | 98    | 117   | 99    | 107   | 91    | 106   | 85    | 75    | 58   |
|                                                                               | PrecMax     | 1692   | 155   | 131   | 291   | 197   | 337   | 215   | 336   | 275   | 275   | 275   | 284   | 219  |
|                                                                               | PrecMin     | 517    | 2     | 6     | 22    | 25    | 11    | 10    | 2     | 7     | 9     | 10    | 7     | 4    |
|                                                                               | PrecStd     | 250    | 34    | 30    | 47    | 45    | 62    | 54    | 67    | 59    | 69    | 59    | 55    | 41   |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 16.2  | 19.2  | 24.5  | 28.4  | 31.3  | 34.2  | 37.4  | 35.8  | 33.6  | 29.4  | 22.8  | 17.7 |
|                                                                               |             | MCWB   | 12.8  | 12.9  | 16.7  | 18.8  | 22.5  | 23.9  | 24.7  | 24.4  | 23.1  | 20.6  | 15.8  | 14.6 |
|                                                                               | 2%          | DB     | 12.4  | 15.6  | 21.6  | 25.9  | 29.0  | 32.4  | 34.2  | 33.2  | 31.5  | 26.4  | 19.9  | 14.1 |
|                                                                               |             | MCWB   | 9.5   | 11.1  | 14.8  | 17.1  | 21.1  | 23.8  | 25.4  | 24.5  | 22.2  | 18.4  | 14.4  | 10.7 |
|                                                                               | 5%          | DB     | 9.1   | 12.3  | 18.2  | 23.5  | 27.4  | 31.0  | 32.6  | 31.6  | 29.5  | 23.9  | 17.3  | 11.0 |
|                                                                               |             | MCWB   | 6.3   | 8.9   | 12.9  | 16.1  | 19.9  | 23.4  | 25.0  | 23.8  | 21.8  | 17.6  | 13.0  | 8.4  |
|                                                                               | 10%         | DB     | 6.1   | 9.0   | 15.5  | 21.1  | 25.5  | 29.2  | 31.0  | 29.8  | 27.4  | 21.6  | 14.8  | 8.3  |
|                                                                               |             | MCWB   | 3.6   | 5.8   | 11.0  | 14.6  | 18.8  | 22.1  | 24.0  | 23.1  | 20.1  | 16.3  | 11.0  | 5.8  |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 13.4  | 14.9  | 17.9  | 20.4  | 23.9  | 26.0  | 27.5  | 27.1  | 24.6  | 22.0  | 17.5  | 15.2 |
|                                                                               |             | MCDB   | 15.5  | 17.6  | 22.9  | 26.1  | 29.4  | 31.5  | 32.9  | 32.5  | 30.6  | 27.7  | 20.7  | 17.4 |
|                                                                               | 2%          | WB     | 10.0  | 11.6  | 15.8  | 18.4  | 22.3  | 24.9  | 26.4  | 25.8  | 23.6  | 20.4  | 15.4  | 11.6 |
|                                                                               |             | MCDB   | 11.9  | 14.4  | 20.1  | 24.1  | 27.5  | 30.5  | 32.3  | 31.0  | 29.3  | 24.3  | 18.6  | 13.8 |
|                                                                               | 5%          | WB     | 6.6   | 8.8   | 13.9  | 17.1  | 21.1  | 24.0  | 25.6  | 24.9  | 22.6  | 18.8  | 13.5  | 8.5  |
|                                                                               |             | MCDB   | 9.1   | 12.5  | 17.7  | 21.9  | 25.8  | 29.4  | 31.2  | 29.9  | 27.7  | 22.4  | 16.6  | 10.3 |
|                                                                               | 10%         | WB     | 3.8   | 6.0   | 11.2  | 15.4  | 20.0  | 23.1  | 24.7  | 23.9  | 21.6  | 17.1  | 11.5  | 6.0  |
|                                                                               |             | MCDB   | 6.0   | 9.1   | 14.7  | 20.4  | 24.4  | 28.2  | 29.7  | 28.3  | 25.9  | 21.0  | 14.5  | 8.2  |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 8.5   | 9.0   | 10.2  | 11.0  | 10.2  | 10.1  | 10.4  | 10.4  | 11.6  | 10.7  | 9.5   | 8.0  |
|                                                                               |             | MCDBR  | 12.2  | 13.2  | 13.4  | 13.7  | 11.7  | 11.4  | 11.7  | 12.2  | 12.8  | 13.0  | 12.3  | 11.1 |
|                                                                               |             | MCWBR  | 9.7   | 9.7   | 8.2   | 7.9   | 6.1   | 5.3   | 4.8   | 5.2   | 5.5   | 6.8   | 8.5   | 8.9  |
|                                                                               | 5% WB       | MCDBR  | 11.3  | 12.8  | 12.2  | 11.8  | 10.5  | 10.3  | 10.6  | 10.5  | 11.0  | 10.5  | 11.2  | 10.4 |
|                                                                               |             | MCWBR  | 9.4   | 10.0  | 8.2   | 7.7   | 6.0   | 5.3   | 5.1   | 5.2   | 5.3   | 6.6   | 8.7   | 9.2  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.295  | 0.317 | 0.340 | 0.390 | 0.431 | 0.429 | 0.450 | 0.443 | 0.395 | 0.352 | 0.303 | 0.289 |      |
|                                                                               | taud        | 2.455  | 2.382 | 2.405 | 2.263 | 2.210 | 2.268 | 2.194 | 2.230 | 2.316 | 2.441 | 2.518 | 2.489 |      |
|                                                                               | Ebn at Noon | 876    | 901   | 915   | 884   | 851   | 851   | 828   | 824   | 843   | 850   | 861   | 856   |      |
|                                                                               | Edh at Noon | 79     | 98    | 107   | 131   | 142   | 134   | 143   | 135   | 116   | 92    | 74    | 71    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1.92   | 2.80  | 3.73  | 4.85  | 5.58  | 6.30  | 6.28  | 5.51  | 4.76  | 3.25  | 2.19  | 1.64  |      |
|                                                                               | RadStd      | 0.22   | 0.26  | 0.27  | 0.36  | 0.42  | 0.56  | 0.38  | 0.30  | 0.31  | 0.36  | 0.19  | 0.20  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +122    | +81     |
| (47) | Regional (32 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +102    | +70     |

Nomenclature: See separate page